

TYRKA, Russian Federation

WMO: 305260

Lat: 54.508N

Lon: 107.138E

Elev: 651

StdP: 93.75

Time Zone: 8.00 (E08)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -43.8	(c) -40.7	(d) -45.1	(e) 0.0	(f) -43.2	(g) -42.3	(h) 0.1	(i) -40.1	(j) 6.9	(k) -23.4	(l) 4.3	(m) -16.2	(n) 0.2	(o) 310	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	28.2	17.1	26.2	16.3	24.4	15.7	18.7	24.5	17.8	23.6	16.8	22.3	1.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	13.1	20.7	15.7	12.1	19.7	14.6	11.2	18.8	56.1	24.1	52.9	23.6	49.7	22.4	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.8	(b) 4.5	(c) 3.7	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 5.8	(b) 4.5	(c) 3.7	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.6	-26.2	-20.7	-11.7	-1.0	6.2	13.4	16.7	14.2	5.9	-3.1	-15.9	-22.6		
	DBStd	15.80	7.68	6.94	6.80	4.83	4.15	3.95	2.62	3.10	4.09	5.27	7.91	8.10		
	HDD10.0	5464	1122	860	673	331	132	13	0	6	133	406	778	1011		
	HDD18.3	8042	1380	1093	932	581	376	153	63	131	372	664	1028	1269		
	CDD10.0	486	0	0	0	0	14	115	209	137	11	0	0	0		
	CDD18.3	22	0	0	0	0	0	5	14	3	0	0	0	0		
	CDH23.3	603	0	0	0	0	15	210	282	91	4	0	0	0		
	CDH26.7	122	0	0	0	0	2	48	58	13	1	0	0	0		
(14) Wind	WSAvg	1.1	0.6	0.7	1.1	1.7	1.9	1.3	1.2	1.0	1.1	1.0	0.8	0.6		
(15) Precipitation	PrecAvg	356	12	8	7	14	27	52	74	73	39	16	17	16		
	PrecMax	436	47	14	23	30	51	109	147	123	85	38	31	46		
	PrecMin	280	3	1	1	4	7	21	39	32	14	2	7	5		
	PrecStd	43	8	3	4	6	11	21	26	28	17	8	5	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.2	-0.3	9.3	17.8	26.3	30.8	30.6	29.1	23.5	14.6	3.0	-3.4		
		MCWB	-7.2	-3.2	2.6	8.0	14.0	17.1	17.8	17.7	13.8	7.2	0.3	-4.5		
	2%	DB	-9.4	-3.6	5.2	13.7	21.7	27.6	28.5	25.9	19.1	10.1	0.1	-6.3		
		MCWB	-10.2	-5.5	0.3	6.0	10.9	16.2	17.4	16.3	11.9	4.6	-1.6	-7.2		
	5%	DB	-11.8	-6.0	2.3	10.7	18.9	25.6	26.6	24.1	15.9	6.8	-2.3	-8.2		
		MCWB	-12.4	-7.5	-1.3	4.3	9.5	15.2	17.3	15.8	10.5	3.0	-3.6	-8.9		
	10%	DB	-14.3	-9.1	-0.2	7.9	16.3	23.6	24.6	21.9	13.3	4.4	-4.6	-10.5		
		MCWB	-14.7	-10.2	-3.0	2.5	8.3	14.7	16.7	15.4	9.1	1.6	-5.6	-11.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.0	-3.0	2.9	8.7	14.1	18.9	20.5	19.5	14.8	7.6	1.0	-4.3		
		MCDB	-6.2	-0.7	8.8	15.9	26.0	25.8	26.0	25.9	21.2	14.1	2.7	-3.4		
	2%	WB	-10.1	-5.3	0.6	6.6	12.0	17.5	19.2	18.0	12.8	5.0	-1.6	-7.0		
		MCDB	-9.4	-3.7	4.6	12.9	19.2	24.9	24.7	23.4	17.9	9.2	0.0	-6.3		
	5%	WB	-12.3	-7.6	-1.2	4.7	10.3	16.2	18.3	16.9	11.1	3.3	-3.6	-8.9		
		MCDB	-11.8	-6.0	2.2	10.1	17.8	23.8	23.9	21.7	15.0	6.5	-2.3	-8.2		
	10%	WB	-14.7	-10.2	-2.8	2.9	8.7	15.0	17.4	15.8	9.5	1.8	-5.6	-11.0		
		MCDB	-14.2	-9.1	-0.3	7.6	15.3	22.4	23.0	20.5	12.8	4.3	-4.6	-10.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.1	17.4	17.9	14.0	13.8	14.5	13.0	12.9	12.0	11.5	11.8	11.5		
		MCDBR	12.5	18.8	19.6	18.8	20.0	19.8	17.8	17.8	17.4	16.4	11.0	9.4		
		MCWBR	11.7	16.6	14.8	11.3	10.6	8.9	8.2	8.6	10.1	11.3	9.2	8.9		
		MCDBR	12.4	18.2	18.5	18.0	18.4	17.2	14.2	13.8	15.6	14.9	11.0	9.5		
	5% WB	MCWBR	11.6	15.8	14.1	11.1	10.2	8.6	7.6	7.7	9.7	10.6	9.2	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.263	0.293	0.316	0.354	0.399	0.440	0.433	0.382	0.335	0.307	0.281	0.259		
	taud		2.066	2.052	2.078	2.204	2.211	2.168	2.229	2.361	2.449	2.413	2.171	2.032		
	Ebn at Noon		661	772	847	867	845	810	807	834	835	774	638	566		
	Edh at Noon		68	99	122	125	132	139	129	106	85	70	61	56		
(44) All-Sky Solar Radiation	RadAvg		1.04	2.09	3.65	4.88	5.24	5.67	5.23	4.44	3.05	1.80	0.99	0.72		
	RadStd		0.08	0.08	0.12	0.33	0.42	0.37	0.50	0.20	0.23	0.22	0.08	0.06		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	-2.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page